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1. Import the attached Netflix csv file in Jupyter notebook and perform following operations using Pandas:

```
In [4]: #a) Print the first 5 rows and last 5 rows of the dataframe
import pandas as pd
fp1=pd.read_csv("Netflix.csv")
fp1.head()
fp1.tail()
pd.concat([fp1.head(),fp1.tail()])
```

```
Out[4]:
```

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thabang Makoa	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabiha Belkacem	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Action & Adventure
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Rishabh Shrivastava	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows

					kanjan Raj, Alam K...								Shows, TV ..
8775	s8803	Movie	Zodiac	David Fincher	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States	November 20, 2019	2007	R	158 min		Cult Movies Dramas Thriller:	
8776	s8804	TV Show	Zombie Dumb	NaN	NaN	NaN	July 1, 2019	2018	TV-Y7	2 Seasons		Kids' TV Korean TV Shows, TV Comedie:	
8777	s8805	Movie	Zombieland	Ruben Fleischer	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States	November 1, 2019	2009	R	88 min		Comedies Horror Movie:	
8778	s8806	Movie	Zoom	Peter Hewitt	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States	January 11, 2020	2006	PG	88 min		Children & Family Movies Comedie:	
8779	s8807	Movie	Zubaan	Mozez Singh	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India	March 2, 2019	2015	TV-14	111 min		Dramas Internationa Movies, Musi & Musical:	



```
In [7]: #b)    Check how many rows and columns are there using Pandas function.
nrows,ncols=fp1.shape
print(nrows,ncols)
print(f"Number of Rows:{nrows}")
print(f"Number of Columns:{ncols}")
```

```
8780 11
Number of Rows:8780
Number of Columns:11
```

```
In [10]: #c)    Print all the column names
fp1.columns
```

```
Out[10]: Index(['show_id', 'type', 'title', 'director', 'cast', 'country', 'date_added',
               'release_year', 'rating', 'duration', 'listed_in'],
              dtype='object')
```

```
In [16]: #d)    Calculate the descriptive statistics of all the variables(integer/float/object etc).
des_status=fp1.describe(include="all")
print(des_status)
```

	show_id	type	title	director	\
count	8780	8780	8780	6150	
unique	8780	2	8780	4516	
top	s1	Movie	Dick Johnson Is Dead	Rajiv Chilaka	
freq	1	6110	1	19	
mean	NaN	NaN	NaN	NaN	
std	NaN	NaN	NaN	NaN	
min	NaN	NaN	NaN	NaN	
25%	NaN	NaN	NaN	NaN	
50%	NaN	NaN	NaN	NaN	
75%	NaN	NaN	NaN	NaN	
max	NaN	NaN	NaN	NaN	

	cast	country	date_added	release_year	\
count	7956	7952	8770	8780.000000	
unique	7671	745	1765	NaN	
top	David Attenborough	United States	January 1, 2020	NaN	
freq	19	2809	108	NaN	
mean	NaN	NaN	NaN	2014.178474	
std	NaN	NaN	NaN	8.827938	
min	NaN	NaN	NaN	1925.000000	
25%	NaN	NaN	NaN	2013.000000	
50%	NaN	NaN	NaN	2017.000000	
75%	NaN	NaN	NaN	2019.000000	
max	NaN	NaN	NaN	2021.000000	

	rating	duration	listed_in
count	8776	8780	8780

```

unique      14      220      514
top    TV-MA  1 Season  Dramas, International Movies
freq      3198      1788      361
mean       NaN       NaN       NaN
std        NaN       NaN       NaN
min        NaN       NaN       NaN
25%        NaN       NaN       NaN
50%        NaN       NaN       NaN
75%        NaN       NaN       NaN
max        NaN       NaN       NaN

```

```

In [15]: des_status2=fp1.describe(include="number")
         print(des_status2)

```

```

         release_year
count      8780.000000
mean       2014.178474
std         8.827938
min       1925.000000
25%       2013.000000
50%       2017.000000
75%       2019.000000
max       2021.000000

```

```

In [14]: des_status1=fp1.describe(include="object")
         print(des_status1)

```

```

         show_id  type      title      director \
count      8780  8780      8780      6150
unique      8780    2      8780      4516
top         s1  Movie  Dick Johnson Is Dead  Rajiv Chilaka
freq         1  6110      1      19

```

```

         cast      country      date_added  rating  duration \
count      7956      7952      8770  8776      8780
unique      7671      745      1765    14      220
top    David Attenborough  United States  January 1, 2020  TV-MA  1 Season
freq         19      2809      108  3198      1788

```

```

         listed_in
count      8780
unique      514
top    Dramas, International Movies
freq      361

```

```
In [23]: #Check the number of unique values for each column
unique_counts=fp1.nunique()
print(unique_counts)
```

```
show_id      8780
type         2
title        8780
director     4516
cast         7671
country       745
date_added   1765
release_year  74
rating        14
duration     220
listed_in    514
dtype: int64
```

```
In [24]: #f) Check the percentage of missing values for each column.
missing_percentage = fp1.isna().mean() * 100
print(missing_percentage)
```

```
show_id      0.000000
type         0.000000
title        0.000000
director     29.954442
cast         9.384966
country      9.430524
date_added   0.113895
release_year  0.000000
rating       0.045558
duration     0.000000
listed_in    0.000000
dtype: float64
```

```
In [26]: #g) Delete all the rows where Director column has missing values.
fp1_cleaned = fp1.dropna(subset=['director'])
# Print the cleaned DataFrame
print(fp1_cleaned)
```

```
   show_id  type      title \
0        s1  Movie  Dick Johnson Is Dead
2        s3  TV Show      Ganglands
5        s6  TV Show      Midnight Mass
```

```

6      s7      Movie  My Little Pony: A New Generation
7      s8      Movie                Sankofa
...      ...      ...
8774    s8802    Movie                Zinzana
8775    s8803    Movie                Zodiac
8777    s8805    Movie                Zombieland
8778    s8806    Movie                Zoom
8779    s8807    Movie                Zubaan

```

```

                                director \
0                                Kirsten Johnson
2                                Julien Leclercq
5                                Mike Flanagan
6      Robert Cullen, José Luis Ucha
7                                Haile Gerima
...                                ...
8774                                Majid Al Ansari
8775                                David Fincher
8777                                Ruben Fleischer
8778                                Peter Hewitt
8779                                Mozez Singh

```

```

                                cast \
0                                NaN
2      Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...
5      Kate Siegel, Zach Gilford, Hamish Linklater, H...
6      Vanessa Hudgens, Kimiko Glenn, James Marsden, ...
7      Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...
...                                ...
8774    Ali Suliman, Saleh Bakri, Yasa, Ali Al-Jabri, ...
8775    Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...
8777    Jesse Eisenberg, Woody Harrelson, Emma Stone, ...
8778    Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...
8779    Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...

```

```

                                country      date_added \
0                                United States  September 25, 2021
2                                NaN            September 24, 2021
5                                NaN            September 24, 2021
6                                NaN            September 24, 2021
7      United States, Ghana, Burkina Faso, United Kin...  September 24, 2021
...                                ...
8774                                United Arab Emirates, Jordan  March 9, 2016
8775                                United States  November 20, 2019
8777                                United States  November 1, 2019
8779                                United States  January 11, 2020

```

8770
8779

UNITED STATES January 11, 2020
India March 2, 2019

```

      release_year rating duration \
0          2020   PG-13    90 min
2          2021   TV-MA    1 Season
5          2021   TV-MA    1 Season
6          2021     PG    91 min
7          1993   TV-MA   125 min
...          ...     ...     ...
8774        2015   TV-MA    96 min
8775        2007     R   158 min
8777        2009     R    88 min
8778        2006     PG    88 min
8779        2015   TV-14   111 min

                                listed_in
0                                Documentaries
2    Crime TV Shows, International TV Shows, TV Act...
5                                TV Dramas, TV Horror, TV Mysteries
6                                Children & Family Movies
7    Dramas, Independent Movies, International Movies
...                                ...
8774        Dramas, International Movies, Thrillers
8775        Cult Movies, Dramas, Thrillers
8777        Comedies, Horror Movies
8778        Children & Family Movies, Comedies
8779    Dramas, International Movies, Music & Musicals

```

[6150 rows x 11 columns]

In [28]:

```

#h)    Print all the records where country has Germany value (including West Germany). If any other country is
filtered_df = fp1[fp1['country'].str.contains('Germany|West Germany', case=False, na=False)]
print(filtered_df)

```

```

      show_id  type      title \
7          s8  Movie      Sankofa
12         s13  Movie      Je Suis Karl
129        s130  Movie  An Unfinished Life
142        s143  Movie  Freedom Writers
172        s173  Movie  School of Rock
...         ...   ...      ...
8590       s8618  Movie      Trash
8634       s8662  Movie  Unfinished Song
8641       s8669  Movie  V for Vendetta

```



```

8702 s8730 Movie Where the Money Is
8718 s8746 Movie Willy Wonka & the Chocolate Factory

director cast \
7 Haile Gerima Kofi Ghanaba, Oyafunmike Ogunlano, Alexandra D...
12 Christian Schwochow Luna Wedler, Jannis Niewöhner, Milan Peschel, ...
129 Lasse Hallström Robert Redford, Jennifer Lopez, Morgan Freeman...
142 Richard LaGravenese Hilary Swank, Patrick Dempsey, Scott Glenn, Im...
172 Richard Linklater Jack Black, Joan Cusack, Mike White, Sarah Sil...
... ...
8590 Stephen Daldry Wagner Moura, Martin Sheen, Rooney Mara, Selto...
8634 Paul Andrew Williams Terence Stamp, Gemma Arterton, Christopher Ecc...
8641 James McTeigue Natalie Portman, Hugo Weaving, Stephen Rea, St...
8702 Marek Kaniévski Paul Newman, Linda Fiorentino, Dermot Mulroney...
8718 Mel Stuart Gene Wilder, Jack Albertson, Peter Ostrum, Roy...

country date_added \
7 United States, Ghana, Burkina Faso, United Kin... September 24, 2021
12 Germany, Czech Republic September 23, 2021
129 Germany, United States September 1, 2021
142 Germany, United States September 1, 2021
172 United States, Germany September 1, 2021
... ...
8590 United Kingdom, Brazil, Germany January 1, 2019
8634 United Kingdom, Germany July 22, 2019
8641 United States, United Kingdom, Germany October 1, 2018
8702 Germany, United States, United Kingdom, Canada January 15, 2020
8718 United States, East Germany, West Germany January 1, 2020

release_year rating duration \
7 1993 TV-MA 125 min
12 2021 TV-MA 127 min
129 2005 PG-13 108 min
142 2007 PG-13 124 min
172 2003 PG-13 110 min
... ...
8590 2014 R 114 min
8634 2012 PG-13 94 min
8641 2005 R 132 min
8702 2000 PG-13 89 min
8718 1971 G 100 min

listed_in
7 Dramas, Independent Movies, International Movies
12 Dramas, International Movies

```

```

129                                     Dramas
142                                     Dramas
172          Comedies, Music & Musicals
...                                     ...
8590          Dramas, Independent Movies, Thrillers
8634          Comedies, Dramas, Independent Movies
8641          Action & Adventure, Dramas, Sci-Fi & Fantasy
8702          Action & Adventure, Comedies, Dramas
8718 Children & Family Movies, Classic Movies, Come...

```

[231 rows x 11 columns]

In [30]:

```

#i)    Expand Duration column into 2 separate columns - First column having the numeric value and other having
#Eg: 3 seasons should be split in 2 columns having 3 in 1st column and seasons in 2nd column.
fp1[['duration_Numeric', 'duration_Unit']] = fp1['duration'].str.extract(r'(\d+)\s*(\w+)')
print(fp1)

```

	show_id	type	title	director \
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson
1	s2	TV Show	Blood & Water	NaN
2	s3	TV Show	Ganglands	Julien Leclercq
3	s4	TV Show	Jailbirds New Orleans	NaN
4	s5	TV Show	Kota Factory	NaN
...	...	...	...	...
8775	s8803	Movie	Zodiac	David Fincher
8776	s8804	TV Show	Zombie Dumb	NaN
8777	s8805	Movie	Zombieland	Ruben Fleischer
8778	s8806	Movie	Zoom	Peter Hewitt
8779	s8807	Movie	Zubaan	Mozez Singh

	cast	country \
0	NaN	United States
1	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa
2	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN
3	NaN	NaN
4	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India
...	...	...
8775	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States
8776	NaN	NaN
8777	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States
8778	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States
8779	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India

date_added release_year rating duration \

```

0    September 25, 2021    2020    PG-13    90 min
1    September 24, 2021    2021    TV-MA    2 Seasons
2    September 24, 2021    2021    TV-MA    1 Season
3    September 24, 2021    2021    TV-MA    1 Season
4    September 24, 2021    2021    TV-MA    2 Seasons
...
8775   November 20, 2019    2007      R    158 min
8776      July 1, 2019    2018   TV-Y7    2 Seasons
8777   November 1, 2019    2009      R     88 min
8778   January 11, 2020    2006     PG     88 min
8779      March 2, 2019    2015   TV-14    111 min

```

```

                                listed_in duration_Numeric \
0                                Documentaries                90
1    International TV Shows, TV Dramas, TV Mysteries          2
2    Crime TV Shows, International TV Shows, TV Act...         1
3                                Docuseries, Reality TV         1
4    International TV Shows, Romantic TV Shows, TV ...         2
...
8775                                Cult Movies, Dramas, Thrillers 158
8776                                Kids' TV, Korean TV Shows, TV Comedies 2
8777                                Comedies, Horror Movies        88
8778                                Children & Family Movies, Comedies 88
8779    Dramas, International Movies, Music & Musicals        111

```

```

duration_Unit
0    min
1    Seasons
2    Season
3    Season
4    Seasons
...
8775    min
8776    Seasons
8777    min
8778    min
8779    min

```

[8780 rows x 13 columns]

```

In [43]: #j) Split Date added into 3 separate columns having date value in 1st column, month value in 2nd column and
if 'date_added' in fp1.columns:
    fp1[['Date', 'Month', 'Year']] = fp1['date_added'].str.split('/', expand=True)
else:
    print("Column 'date_added' not found in the DataFrame. Check previous operations.")

```

```
print(fp1)
```

Column 'date_added' not found in the DataFrame. Check previous operations.

	show_id	type	title	director \
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson
1	s2	TV Show	Blood & Water	NaN
2	s3	TV Show	Ganglands	Julien Leclercq
3	s4	TV Show	Jailbirds New Orleans	NaN
4	s5	TV Show	Kota Factory	NaN
...	...	...	...	...
8775	s8803	Movie	Zodiac	David Fincher
8776	s8804	TV Show	Zombie Dumb	NaN
8777	s8805	Movie	Zombieland	Ruben Fleischer
8778	s8806	Movie	Zoom	Peter Hewitt
8779	s8807	Movie	Zubaan	Mozez Singh

	cast	country \
0	NaN	United States
1	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa
2	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN
3	NaN	NaN
4	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India
...	...	...
8775	Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...	United States
8776	NaN	NaN
8777	Jesse Eisenberg, Woody Harrelson, Emma Stone, ...	United States
8778	Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...	United States
8779	Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...	India

	release_year	rating	duration \
0	2020	PG-13	90 min
1	2021	TV-MA	2 Seasons
2	2021	TV-MA	1 Season
3	2021	TV-MA	1 Season
4	2021	TV-MA	2 Seasons
...	...	...	...
8775	2007	R	158 min
8776	2018	TV-Y7	2 Seasons
8777	2009	R	88 min
8778	2006	PG	88 min
8779	2015	TV-14	111 min

	listed_in	duration_Numeric \
0	Documentaries	90
1	International TV Shows. TV Dramas. TV Mvseries	2

```

2    Crime TV Shows, International TV Shows, TV Act...      1
3                                Docuseries, Reality TV      1
4    International TV Shows, Romantic TV Shows, TV ...      2
...
8775                                Cult Movies, Thrillers    158
8776    Kids' TV, Korean TV Shows, TV Comedies              2
8777                                Comedies, Horror Movies    88
8778                                Children & Family Movies    88
8779    Dramas, International Movies, Music & Musicals      111

```

```

duration_Unit  Date  Month  Year
0             min  25.0   9.0  2021.0
1         Seasons  24.0   9.0  2021.0
2         Season  24.0   9.0  2021.0
3         Season  24.0   9.0  2021.0
4         Seasons  24.0   9.0  2021.0
...
8775             min  20.0  11.0  2019.0
8776         Seasons   1.0   7.0  2019.0
8777             min   1.0  11.0  2019.0
8778             min  11.0   1.0  2020.0
8779             min   2.0   3.0  2019.0

```

[8780 rows x 15 columns]

In [44]:

```

#k)    Print the number of TV shows/Movies released in each year.
release_counts = fp1.groupby(['release_year', 'type']).size().unstack(fill_value=0)

# Print the result
print(release_counts)

```

```

type      Movie  TV Show
release_year
1925         0         1
1942         2         0
1943         3         0
1944         3         0
1945         3         1
...
2017       765       265
2018       764       377
2019       628       396
2020       517       435
2021       277       315

```

[74 rows x 2 columns]

In [45]:

```
#L)    Rename the column title with movie_title
fp1.rename(columns={'title': 'movie_title'}, inplace=True)
print(fp1.head())
```

	show_id	type	movie_title	director	\
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	
1	s2	TV Show	Blood & Water	NaN	
2	s3	TV Show	Ganglands	Julien Leclercq	
3	s4	TV Show	Jailbirds New Orleans	NaN	
4	s5	TV Show	Kota Factory	NaN	

	cast	country	\
0	NaN	United States	
1	Ama Qamata, Khosi Ngema, Gail Mablane, Thaban...	South Africa	
2	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	
3	NaN	NaN	
4	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	

	release_year	rating	duration	\
0	2020	PG-13	90 min	
1	2021	TV-MA	2 Seasons	
2	2021	TV-MA	1 Season	
3	2021	TV-MA	1 Season	
4	2021	TV-MA	2 Seasons	

	listed_in	duration_Numeric	\
0	Documentaries	90	
1	International TV Shows, TV Dramas, TV Mysteries	2	
2	Crime TV Shows, International TV Shows, TV Act...	1	
3	Docuseries, Reality TV	1	
4	International TV Shows, Romantic TV Shows, TV ...	2	

	duration_Unit	Date	Month	Year
0	min	25.0	9.0	2021.0
1	Seasons	24.0	9.0	2021.0
2	Season	24.0	9.0	2021.0
3	Season	24.0	9.0	2021.0
4	Seasons	24.0	9.0	2021.0

In [46]:

```
#m)    Split Listed_in column into 3 different columns with col name (Genre1, Genre2, Genre3). Split the column
fp1[['Genre1', 'Genre2', 'Genre3']] = fp1['listed_in'].str.split(',', expand=True)
print(fp1)
```

```

show_id  type  movie_title  director \
0      s1  Movie  Dick Johnson Is Dead  Kirsten Johnson
1      s2  TV Show  Blood & Water  NaN
2      s3  TV Show  Ganglands  Julien Leclercq
3      s4  TV Show  Jailbirds New Orleans  NaN
4      s5  TV Show  Kota Factory  NaN
...      ...      ...      ...      ...
8775  s8803  Movie  Zodiac  David Fincher
8776  s8804  TV Show  Zombie Dumb  NaN
8777  s8805  Movie  Zombieland  Ruben Fleischer
8778  s8806  Movie  Zoom  Peter Hewitt
8779  s8807  Movie  Zubaan  Mozez Singh

cast  country \
0      NaN  United States
1  Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...  South Africa
2  Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...  NaN
3      NaN  NaN
4  Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...  India
...      ...      ...
8775  Mark Ruffalo, Jake Gyllenhaal, Robert Downey J...  United States
8776      NaN  NaN
8777  Jesse Eisenberg, Woody Harrelson, Emma Stone, ...  United States
8778  Tim Allen, Courteney Cox, Chevy Chase, Kate Ma...  United States
8779  Vicky Kaushal, Sarah-Jane Dias, Raaghav Chanan...  India

release_year  rating  duration \
0      2020  PG-13  90 min
1      2021  TV-MA  2 Seasons
2      2021  TV-MA  1 Season
3      2021  TV-MA  1 Season
4      2021  TV-MA  2 Seasons
...      ...      ...      ...
8775      2007  R  158 min
8776      2018  TV-Y7  2 Seasons
8777      2009  R  88 min
8778      2006  PG  88 min
8779      2015  TV-14  111 min

listed_in  duration_Numeric \
0      Documentaries  90
1  International TV Shows, TV Dramas, TV Mysteries  2
2  Crime TV Shows, International TV Shows, TV Act...  1
3      Docuseries, Reality TV  1

```

```

4      International TV Shows, Romantic TV Shows, TV ...      2
...      ...      ...
8775      Cult Movies, Dramas, Thrillers      158
8776      Kids' TV, Korean TV Shows, TV Comedies      2
8777      Comedies, Horror Movies      88
8778      Children & Family Movies, Comedies      88
8779      Dramas, International Movies, Music & Musicals      111

```

```

      duration_Unit  Date  Month   Year      Genre1 \
0      min      25.0    9.0   2021.0      Documentaries
1      Seasons      24.0    9.0   2021.0      International TV Shows
2      Season      24.0    9.0   2021.0      Crime TV Shows
3      Season      24.0    9.0   2021.0      Docuseries
4      Seasons      24.0    9.0   2021.0      International TV Shows
...      ...      ...      ...      ...
8775      min      20.0   11.0   2019.0      Cult Movies
8776      Seasons      1.0    7.0   2019.0      Kids' TV
8777      min      1.0   11.0   2019.0      Comedies
8778      min      11.0    1.0   2020.0      Children & Family Movies
8779      min      2.0    3.0   2019.0      Dramas

```

```

      Genre2      Genre3
0      None      None
1      TV Dramas      TV Mysteries
2      International TV Shows      TV Action & Adventure
3      Reality TV      None
4      Romantic TV Shows      TV Comedies
...      ...      ...
8775      Dramas      Thrillers
8776      Korean TV Shows      TV Comedies
8777      Horror Movies      None
8778      Comedies      None
8779      International Movies      Music & Musicals

```

[8780 rows x 18 columns]

2. Import both the attached files (student.csv and marks.csv) in Jupyter notebook and perform following operations:

In [75]:

```

#a)      Combine both the dataframes into single dataframe which will have all the records from both the tables.
import pandas as pd
df1=pd.read_csv("Students.csv")
df2=pd.read_csv("marks.csv")
#combined_df = pd.concat([df1, df2], ignore_index=True)
combined_df = pd.concat([df1, df2])
print(combined_df)

```



```
print(combined_df)
```

	Student_id	Age	Gender	Grade	Employed	Mark	City
0	1	19.0	Male	1st Class	yes	NaN	NaN
1	2	20.0	Female	2nd Class	no	NaN	NaN
2	3	18.0	Male	1st Class	no	NaN	NaN
3	4	21.0	Female	2nd Class	no	NaN	NaN
4	5	19.0	Male	1st Class	no	NaN	NaN
..	...	...	...	...	...	...	...
224	228	NaN	NaN	NaN	NaN	99.0	Pune
225	229	NaN	NaN	NaN	NaN	70.0	Chennai
226	230	NaN	NaN	NaN	NaN	55.0	Delhi
227	231	NaN	NaN	NaN	NaN	97.0	Mumbai
228	232	NaN	NaN	NaN	NaN	59.0	Pune

[461 rows x 7 columns]

```
In [73]: #b) Print the maximum and minimum marks Gender wise.
#combined_df = combined_df.dropna(subset=['Mark'])
#print(combined_df)
gender_wise_marks = combined_df.groupby('Gender')['Mark'].agg(['max', 'min'])
print(gender_wise_marks)
```

Empty DataFrame

Columns: [max, min]

Index: []

```
In [68]: #c) Print all the students IDs and their marks who have scored more than the average marks of the class.
average_marks = combined_df['Mark'].mean()

# Filter students who scored more than the average marks
above_average_students = combined_df[combined_df['Mark'] > average_marks]

# Print Student_ID and Marks for these students
print(above_average_students[['Student_id', 'Mark']])
```

	Student_id	Mark
0	1	95.0
2	3	98.0
3	4	75.0
4	5	89.0
9	12	90.0
..	...	...
218	222	74.0
219	223	79.0

```

221         225  72.0
224         228  99.0
227         231  97.0

```

[137 rows x 2 columns]

```

In [69]: #d)    Print the dataframe who are Males and are Employed.
         filtered_df = combined_df[(combined_df['Gender'] == 'Male') & (combined_df['Employed'] == 'Employed')]

         # Print the filtered DataFrame
         print(filtered_df)

```

Empty DataFrame

Columns: [Student_id, Age, Gender, Grade, Employed, Mark, City]

Index: []

```

In [70]: #e)    Create a new Column 'IQ_level' which will have 3 values (Intelligent, Mediocre, weak). If student scored
         #if student scored between 50-80, then Mediocre, else weak.
         bins = [0, 50, 80, float('inf')]
         labels = ['Weak', 'Mediocre', 'Intelligent']

         # Create the new column 'IQ_level' based on the marks
         combined_df['IQ_level'] = pd.cut(combined_df['Mark'], bins=bins, labels=labels, right=False)

         # Print the updated DataFrame
         print(combined_df)

```

	Student_id	Age	Gender	Grade	Employed	Mark	City	IQ_level
0	1	NaN	NaN	NaN	NaN	95.0	Chennai	Intelligent
1	2	NaN	NaN	NaN	NaN	70.0	Delhi	Mediocre
2	3	NaN	NaN	NaN	NaN	98.0	Mumbai	Intelligent
3	4	NaN	NaN	NaN	NaN	75.0	Pune	Mediocre
4	5	NaN	NaN	NaN	NaN	89.0	Kochi	Intelligent
..	...	...	...	...	...	...	...	...
224	228	NaN	NaN	NaN	NaN	99.0	Pune	Intelligent
225	229	NaN	NaN	NaN	NaN	70.0	Chennai	Mediocre
226	230	NaN	NaN	NaN	NaN	55.0	Delhi	Mediocre
227	231	NaN	NaN	NaN	NaN	97.0	Mumbai	Intelligent
228	232	NaN	NaN	NaN	NaN	59.0	Pune	Mediocre

[229 rows x 8 columns]

```

In [71]: #f)    Count the number of males and females from each city

```

```

""" Count the number of males and females from each city.
gender_city_counts = combined_df.groupby(['City', 'Gender']).size().unstack(fill_value=0)

# Print the result
print(gender_city_counts)

```

Empty DataFrame

Columns: []

Index: []

In [76]:

```

#g) Print the top 5 Male scorers.
male_students = combined_df[combined_df['Gender'] == 'Male']

# Sort by 'Marks' in descending order
top_male_scorers = male_students.sort_values(by='Mark', ascending=False).head(5)

# Print the top 5 male scorers
print(top_male_scorers)

```

	Student_id	Age	Gender	Grade	Employed	Mark	City
0	1	19.0	Male	1st Class	yes	NaN	NaN
2	3	18.0	Male	1st Class	no	NaN	NaN
4	5	19.0	Male	1st Class	no	NaN	NaN
5	6	20.0	Male	2nd Class	yes	NaN	NaN
7	8	21.0	Male	3rd Class	yes	NaN	NaN

In [77]:

```

#h) Replace the Male value with M and Female value with F and export this dataframe to excel file in D: (D:\test.csv)

# Replace 'Male' with 'M' and 'Female' with 'F'
combined_df['Gender'] = combined_df['Gender'].replace({'Male': 'M', 'Female': 'F'})

# Specify the path where you want to save the file
output_path = r'D:\test.csv'

# Export the DataFrame to a CSV file
combined_df.to_csv(output_path, index=False)

print(f"File has been saved to {output_path}")

```

File has been saved to D:\test.csv

In [79]:

```

#i) Check if any student_ID is duplicated.

```

```

duplicates = combined_df[combined_df.duplicated('Student_id')]

if not duplicates.empty:
    print("Duplicated student_IDs found:")
    print(duplicates)
else:
    print("No duplicated student_IDs found.")

```

Duplicated student_IDs found:

	Student_id	Age	Gender	Grade	Employed	Mark	City
0	1	NaN	NaN	NaN	NaN	95.0	Chennai
1	2	NaN	NaN	NaN	NaN	70.0	Delhi
2	3	NaN	NaN	NaN	NaN	98.0	Mumbai
3	4	NaN	NaN	NaN	NaN	75.0	Pune
4	5	NaN	NaN	NaN	NaN	89.0	Kochi
..	...	...	...	...	...	...	...
224	228	NaN	NaN	NaN	NaN	99.0	Pune
225	229	NaN	NaN	NaN	NaN	70.0	Chennai
226	230	NaN	NaN	NaN	NaN	55.0	Delhi
227	231	NaN	NaN	NaN	NaN	97.0	Mumbai
228	232	NaN	NaN	NaN	NaN	59.0	Pune

[229 rows x 7 columns]

```

In [80]: #j) Create a separate dataframe which will have all the Integer/Float variables.
numeric_df = combined_df.select_dtypes(include=['int64', 'float64'])

print("DataFrame with only Integer/Float variables:")
print(numeric_df)

```

DataFrame with only Integer/Float variables:

	Student_id	Age	Mark
0	1	19.0	NaN
1	2	20.0	NaN
2	3	18.0	NaN
3	4	21.0	NaN
4	5	19.0	NaN
..	...	...	...
224	228	NaN	99.0
225	229	NaN	70.0
226	230	NaN	55.0
227	231	NaN	97.0
228	232	NaN	59.0

[461 rows x 3 columns]

In [83]:

```
#k) Get those Student_IDs which are present in Students table but not in Marks table.
missing_marks = df1[~df1['Student_id'].isin(df2['Student_id'])] # Changed 'Students' to 'df1' and 'marks' to 'df2'
print("student_IDs present in Students table but not in Marks table:")
print(missing_marks)
```

student_IDs present in Students table but not in Marks table:

	Student_id	Age	Gender	Grade	Employed
8	9	22	Female	3rd Class	yes
9	10	21	Male	1st Class	no
48	49	19	Male	2nd Class	no

In []:

3. Explain the concept of missing values. How can you identify the missing values in a Pandas DataFrame ? Missing values refer to the absence of data in a dataset. They can occur due to various reasons, such as: Data entry errors, Incomplete data collection Corruption of data Irrelevant or non-applicable fields In datasets, missing values are often represented as: NaN (Not a Number) in numerical columns None in object (string) columns Blank spaces or placeholders like NA, NULL, or ? Handling missing values is crucial because they can lead to incorrect analysis, biased results, or errors during data processing. Identifying Missing Values in a Pandas DataFrame Pandas provides several methods to identify missing values in a DataFrame. Here are the most common ones:

1. isna() or isnull() These methods return a DataFrame of the same shape as the original, with True for missing values and False for non-missing values.
2. notna() or notnull() These methods are the opposite of isna() and return True for non-missing values and False for missing values.

4.What are the different ways of treating/Imputing/Deleting the missing values. Explain with example. **bold text

Handling missing values is a critical step in data preprocessing. Depending on the nature of the data and the analysis, you can choose to treat, impute, or delete missing values. Below are the **common** approaches with examples: python